Progress Log

Wi-Fi-Controlled Unmanned Ground Vehicle

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Assigned by
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Acknowledgment

We would like to thank Prof. Dr. GVV Sharma for assigning this project and giving us the opportunity to work on a practical wireless-control system.

The project involved combining circuit design, embedded programming, motor control, and wireless communication. The practical work helped us understand the challenges involved in developing a reliable physical system.

This document records the models explored, the progress made during development, the problems encountered, and the possible future extensions of the project.

Photographs, circuit diagrams, interface screenshots, source-code locations, and demonstration links are included wherever applicable.

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1 Progress Log

This section records the development of the UGV in chronological order. It describes the work carried out during each stage, the hardware and software used, the problems encountered, and the changes made during development.

The detailed technical descriptions of the models are provided separately in Section 2. Clickable references are included at the end of each day's entry.

1.1 Day 1: Component Collection and Bluetooth Control

1.1.1 Component collection

We began by collecting the main components required to build the UGV. These included the chassis, two DC gear motors, an L293D motor driver, an ESP32 DevKit, connecting wires, and a battery supply.

We reviewed reference material to understand the general arrangement of the controller, motor driver, motors, and power supply.

We also prepared a circuit representation in Tinkercad to plan the connections before proceeding with the physical assembly.

The reference material included an L298N motor-driver example, while our actual implementation used the L293D driver. The reference was therefore used as a general guide, with the actual connections checked against our hardware.

1.1.2 First Bluetooth implementation

Our first objective was to establish wireless motor control using an ESP32 DevKit.

Power Supply and Distribution. The ESP32 Bluetooth-controlled UGV was powered using a two-cell lithium-ion battery pack with an operating voltage ranging from 7.4 V to 8.4 V. An LM2596 DC–DC buck converter was used to step down the battery voltage to a regulated 5 V supply. This regulated output powered the logic supply (VCC1, pin 16) of the L293D motor driver and the ESP32 development board through its designated 5 V input. Meanwhile, the battery voltage was connected directly to the motor supply input (VCC2, pin 8) of the L293D to provide the required power for the DC motors. The grounds of the battery, buck converter, ESP32, and L293D were connected together to establish a common electrical reference for the control signals.

We selected the Dabble mobile application because it provides a Gamepad interface that communicates with the ESP32 through Bluetooth.

The ESP32 was configured with the Bluetooth device name `ESP32_RC_Car`. The user connects the phone to the ESP32, opens the Gamepad interface, and uses the directional controls to operate the motors.

The ESP32 processes the received inputs and generates the direction and PWM signals required by the L293D motor driver.

The initial Bluetooth implementation successfully demonstrated wireless control of the rover.

1.1.3 Problems encountered

A few practical problems appeared during the initial assembly and testing.

- **Improper motor soldering:** Some motor connections were not soldered properly. This was resolved by reworking and resoldering the affected connections.
- **Left and Right Turning Calibration.** During the initial testing of the Bluetooth-controlled UGV, difficulties were encountered in achieving consistent left and right turning movements. The two DC gear motors exhibited differences in their rotational response, and their physical orientations required appropriate direction mapping in the firmware. Consequently, the initial turning commands did not always produce the intended movement. This was addressed by applying suitable speed trimming to improve the coordination between the two motors and obtain more predictable turning behaviour.
- **Loose jumper-wire connections:** Several connections were not sufficiently secure, causing intermittent contact. This remained a recurring issue during later testing.
- **Power stability:** During later Wi-Fi experiments, the ESP32 did not behave reliably. We suspected that the power arrangement might have contributed, but did not establish a single confirmed cause.

1.1.4 Outcome

The ESP32 DevKit successfully operated the initial Bluetooth-controlled rover.

This model established the basic wireless motor-control approach that was later extended through browser-based Wi-Fi control.

The detailed implementation is described in Model 1: Bluetooth Control Using Dabble (Section 2.1).

1.2 Day 2: Local Wi-Fi Joystick Control (Main Project)

1.2.1 Transition to Wi-Fi control

After developing the Bluetooth-controlled model, we wanted to explore a browser-based interface for operating the rover.

The objective was to create a local Wi-Fi network using the microcontroller itself. A phone or laptop could connect to this network and control the rover through a webpage.

We initially attempted an ESP32-based Wi-Fi setup. However, its operation was not sufficiently reliable during testing. We also experimented with a NodeMCU board, but the issue persisted.

To continue development, we shifted the local Wi-Fi implementation to a Raspberry Pi Pico W.

1.2.2 Pico W local network

The Raspberry Pi Pico W was configured to create its own wireless access point with the network name `UGV`.

A phone or laptop can connect directly to this network without requiring an external router or campus Wi-Fi.

After connecting, the user opens the control webpage through the browser at:

```
http://192.168.4.1
```

The webpage provides a virtual joystick. Its horizontal and vertical values are transmitted to the Pico W through HTTP requests.

The controller processes these values and generates the corresponding motor commands using differential drive.

1.2.3 Multi-device connectivity

One of the important advantages of the local Wi-Fi architecture is that the rover does not have to be controlled from only one phone or laptop.

The Pico W access point supports up to four connected Wi-Fi clients simultaneously. Therefore, multiple devices can join the `UGV` network and access the browser-based control interface at the same time.

This provides a useful multi-device capability:

- Multiple phones or laptops can connect to the rover's local network.
- Each connected device can open the same browser-based control webpage.
- The control interface does not need to be installed separately on every device.
- The rover can be accessed locally without an internet connection or an external router.

The maximum supported access-point client count is **4 devices simultaneously**, according to the Pico W wireless access-point specification.

Since all connected devices communicate with the same rover, they share the same motor-control system. Commands received from different devices are processed by the same controller; a later command can therefore change the movement requested by an earlier device.

This is shared multi-device access, rather than independent control of four separate vehicles.

The four-device limit refers to Wi-Fi client connections, not a guarantee that four simultaneous high-frequency joystick streams will have identical response performance.

1.2.4 Continuous Joystick Control and Motor Calibration.

Unlike the Bluetooth button-based implementation, the Wi-Fi-controlled UGV uses a continuous joystick interface, where the horizontal and vertical joystick coordinates are continuously translated into individual motor commands. During testing, it was observed that additional motor speed trimming was not necessary for the joystick-based implementation, as the continuous variation of the motor commands provided smoother control over the rover's movement and turning behaviour. This made the control response more consistent during directional transitions without requiring the manual speed-trimming adjustments used in the Bluetooth model.

1.2.5 Joystick control and stopping behaviour

The browser interface provides proportional control rather than only fixed directional commands.

The joystick generates two normalised values:

- x: horizontal movement, representing left and right steering.
- y: vertical movement, representing forward and reverse movement.

The Pico W combines these values to calculate the left and right motor commands.

A STOP command is also provided. The firmware uses a watchdog mechanism to stop the motors if fresh commands are no longer received.

The browser sends joystick updates at an interval of 20 milliseconds, corresponding to a nominal update frequency of 50 Hz.

The watchdog timeout in the documented firmware is 250 milliseconds.

1.2.6 Campus Wi-Fi distinction

The Pico W local access-point model is independent of the campus network. It creates its own network for direct communication between the phone and rover.

The IITH campus network uses WPA2-Enterprise authentication. The local Pico W implementation was not used as a successful campus-network connection.

This local browser-control model is therefore separate from the ESP32-CAM campus-network model.

1.2.7 Outcome

The Raspberry Pi Pico W successfully provided the local access-point architecture required for browser-based control.

The model introduced:

- Local Wi-Fi access point named `UGV`.
- Browser-based joystick interface.
- HTTP communication with the controller.
- Proportional differential-drive control.
- PWM-based motor speed control.
- Multi-device access for up to four Wi-Fi clients.
- STOP command and watchdog-based stopping.

The detailed architecture and operation are explained in Model 2: Local Wi-Fi Joystick Control Using Raspberry Pi Pico W (Section 2.2).

Before the Day 3 implementation log, it is worth noting that the ESP32-CAM model was developed almost entirely at the theoretical and design level, including its system architecture, circuit connections, and proposed Wi-Fi-based camera functionality. However, due to component constraints, we were unable to produce the physical model or demonstrate its operation. The theoretical development and implementation plan were completed, providing a foundation for its future physical realization.

1.3 Day 3: ESP32-CAM Campus Camera Rover

1.3.1 Camera-assisted rover concept

The third stage involved exploring a camera-assisted rover using an ESP32-CAM board.

The intention was to combine remote rover operation with a live camera feed. This would allow an operator to observe the surroundings of the rover while controlling its movement.

Unlike the local Pico W access-point model, this approach was intended to use the IITH campus wireless network.

1.3.2 Campus network connection

We successfully established the ESP32-CAM connection to the IITH campus Wi-Fi network using WPA2-Enterprise authentication.

This confirmed that the campus-network connection could be established with the ESP32-CAM.

The remaining difficulty was associated with the camera hardware and interfacing.

1.3.3 Camera interfacing problem

During camera setup, we encountered a hardware or camera-interfacing problem that prevented successful camera initialisation.

Although the campus Wi-Fi connection was established, the camera problem meant that we could not complete the intended live-streaming demonstration.

The two outcomes are therefore distinct:

- ESP32-CAM campus Wi-Fi connection using WPA2-Enterprise: successful.
- Camera initialisation and live streaming: incomplete due to the camera interfacing problem.

1.3.4 Further work

The next step would be to verify the camera hardware independently, establish a stable video stream, and then integrate it with the rover's control interface.

The intended camera rover architecture and the interfacing issue are discussed in Model 3: ESP32-CAM Campus Camera Rover (Section 2.3).

2 Project Models

This section describes the three main models explored during the project. Each model represents a different approach to wireless control or remote operation of the UGV.

The models are presented according to their intended architecture and technical operation. Their practical development history is recorded separately in the Progress Log.

2.1 Model 1: Bluetooth Control Using Dabble

2.1.1 Objective and system overview

The first model was developed to establish wireless control of the UGV through a mobile phone.

An ESP32 DevKit was used as the main controller, while the Dabble mobile application provided the user interface.

The application communicates with the ESP32 through Bluetooth and supplies directional and speed-level inputs.

The ESP32 processes these inputs and generates the required direction and PWM signals for the L293D motor driver.

The overall control sequence is:

1. The phone connects to the ESP32 through Bluetooth.
2. The user opens the Gamepad module in Dabble.
3. Directional and speed-level inputs are read by the ESP32.
4. The controller calculates the required motor commands.
5. The L293D operates the motors according to the received signals.

2.1.2 Bluetooth connection and Dabble interface

The Bluetooth device name configured in the firmware is `ESP32_RC_Car`.

The Dabble application provides a Gamepad interface through which the user sends directional commands. The ESP32 reads the relevant button states and translates them into motor-control values.

The phone supplies the user's input, while the ESP32 performs the control calculations and generates the signals required by the motor driver.

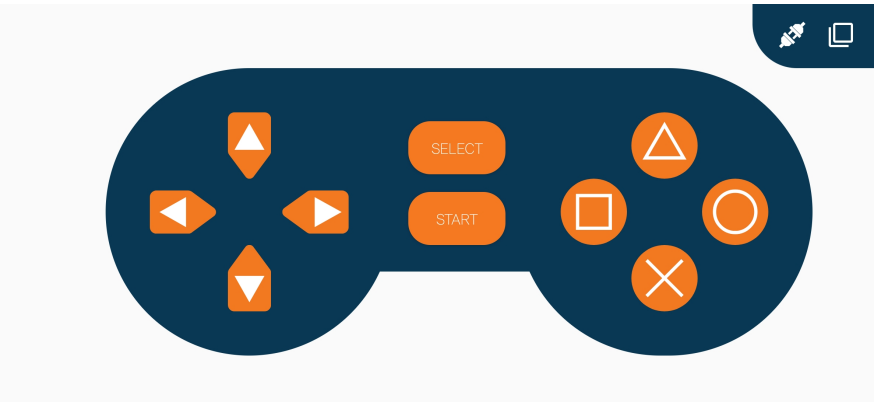


Figure 1: Dabble Gamepad interface used to control the rover.

2.1.3 Motor driver and GPIO mapping

The Bluetooth model uses an L293D motor driver.

The left and right motors have separate direction inputs and enable signals. The enable pins are used for PWM-based speed control, while the direction inputs determine the rotation direction.

Function	ESP32 GPIO	L293D pin
Left direction input 1	32	2
Left direction input 2	33	7
Left enable / PWM	27	1
Right direction input 1	25	10
Right direction input 2	26	15
Right enable / PWM	14	9

The right motor is configured with an inversion setting to account for its physical orientation. A small right-side trim is also used to compensate for differences in motor response.

2.1.4 Circuit diagram

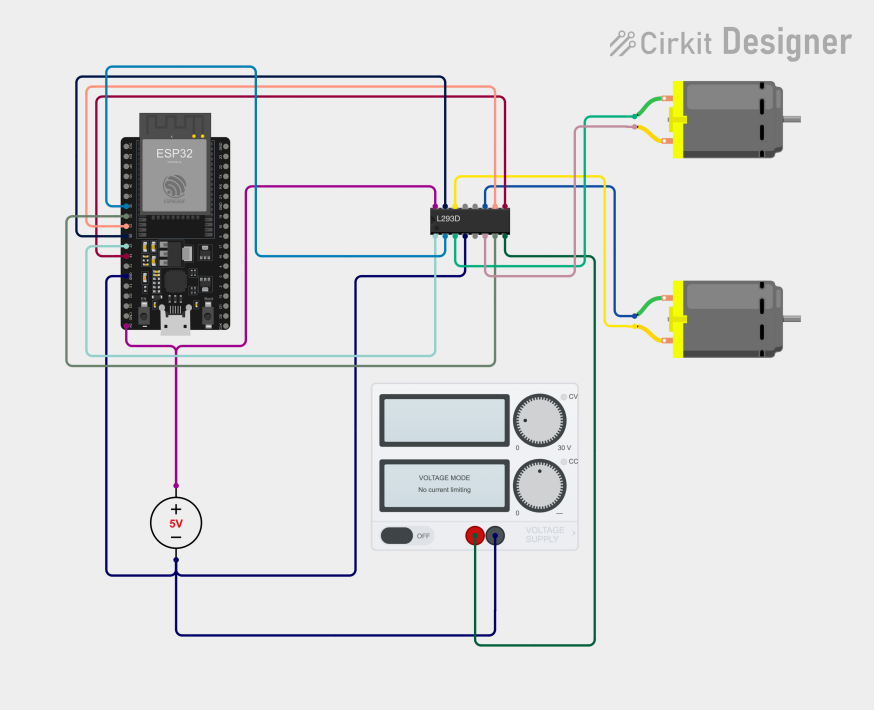


Figure 2: ESP32 Bluetooth/Dabble model circuit diagram.

2.1.5 PWM speed control

The firmware uses PWM to control the effective drive supplied to the motors.

The configured PWM parameters are:

- PWM frequency: 5 kHz.
- Resolution: 8 bits.
- PWM range: 0–255.
- Initial speed level: 5.
- Available speed levels: 3–7.
- Minimum non-zero PWM value: 90.
- Maximum PWM value: 255.

The Triangle button increases the selected speed level, while the Cross button decreases it.

The minimum non-zero PWM value helps the motors overcome their starting friction. Actual speed also depends on the motor, battery condition, mechanical load, and surface.

2.1.6 Model photographs and demonstration

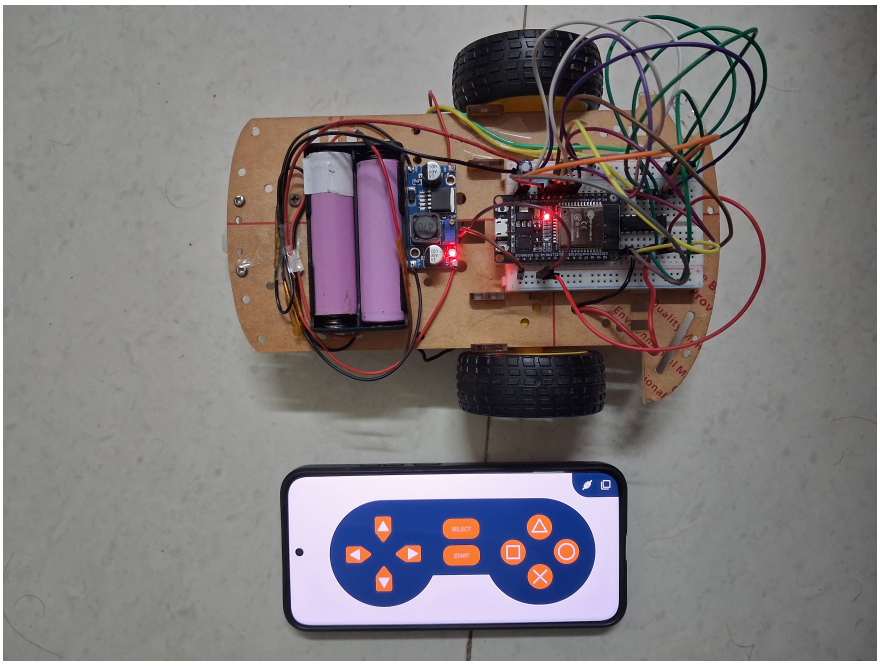


Figure 3: Assembled Bluetooth-controlled rover.

2.1.7 Source code and demonstration

The complete firmware is maintained separately from this report.

Source-code path:

```
codes/bluetooth/main.ino
```

Circuit diagram link:

Open ESP32 Bluetooth/Dabble circuit in Cirkit Designer

Demonstration video:

Watch Bluetooth/Dabble rover demonstration

2.1.8 Model outcome

The Bluetooth model established the initial wireless motor-control approach. The ESP32 DevKit communicated with the Dabble application and used the received inputs to control the two motors.

The implementation introduced the basic control functions required by the later models: direction selection, PWM speed control, and motor inversion.

2.2 Model 2: Local Wi-Fi Joystick Control Using Raspberry Pi Pico W

2.2.1 Objective and system overview

The second model was developed to provide wireless control through a browser-based joystick interface.

A Raspberry Pi Pico W acts as both the controller and the local Wi-Fi access point. A phone or laptop connects directly to the network created by the Pico W and opens the control webpage in a browser.

The webpage sends joystick coordinates to the controller using HTTP requests. The Pico W processes these values and generates the corresponding motor commands.

This approach avoids the need for a separate mobile application and allows the control interface to be accessed through a standard browser.

2.2.2 Local Wi-Fi access point

The Pico W creates a wireless access point with the SSID:

`UGV`

The phone or laptop connects directly to this network.

The Pico W runs an HTTP server on port 80. After connecting to the access point, the user opens the following address in the browser:

`http://192.168.4.1`

This is a local network connection. It does not require an external router or an internet connection.

2.2.3 Multi-device access and simultaneous connectivity

A major feature of this architecture is its ability to support multiple connected devices.

The Raspberry Pi Pico W supports up to **4 simultaneous Wi-Fi clients** in access-point mode. This means that up to four phones or laptops can connect to the same `UGV` wireless network at the same time.

Each connected device can independently open the browser-based control webpage using the local IP address.

This makes the system useful in situations where more than one person needs access to the rover's interface.

For example, one device can be used to operate the joystick while other connected devices can open the same interface to observe the control page and the system's response.

The multi-device architecture provides the following advantages:

- **Multiple client connections:** Up to four devices can connect to the rover's access point simultaneously.
- **Shared browser interface:** Each connected device can access the same control webpage without installing a dedicated application.
- **Flexible access:** Different phones and laptops can connect to the rover using the same local network.
- **No external router required:** The Pico W itself creates the network, making the system self-contained.
- **Local operation:** Devices can access the rover even when an internet connection is unavailable.

The four-client limit is the maximum supported simultaneous access-point connection count, not a guarantee of identical performance for every connected device.

All devices communicate with the same controller and therefore share the same motor outputs. If multiple devices issue movement commands, the controller processes the received requests as they arrive. A later command may override an earlier movement command.

Thus, the system provides simultaneous multi-device access and shared control, rather than four independent motor-control channels.

The four-client capability is specified in the Raspberry Pi Pico W wireless documentation and the Arduino-Pico Wi-Fi documentation.

2.2.4 Browser-based joystick interface

The webpage contains a virtual joystick that can be operated using a touchscreen or a mouse.

The joystick provides two normalised values:

$$-1 \leq x \leq 1$$

$$-1 \leq y \leq 1$$

The horizontal coordinate x represents steering, while the vertical coordinate y represents forward and reverse movement.

The interface also provides a STOP button.

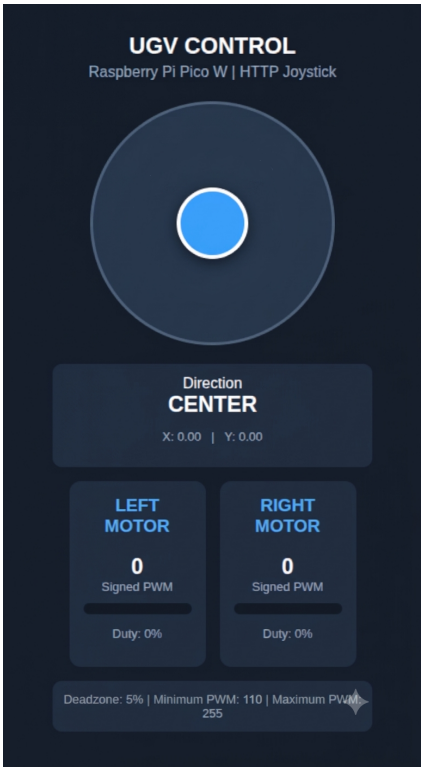


Figure 4: Browser-based local Wi-Fi joystick interface.

2.2.5 HTTP communication

The browser communicates with the Pico W through HTTP requests.

When the joystick is moved, the webpage sends the horizontal and vertical values through a request following this general format:

```
/drive?x=VALUE&y=VALUE
```

The controller reads the request, extracts the two coordinates, and passes them to the motor-control function.

A separate request is used to stop the rover:

```
/stop
```

The server returns a response after processing the request.

The webpage and motor-control firmware therefore operate as separate parts of the system. The webpage supplies the requested movement, while the Pico W calculates and applies the motor commands.

2.2.6 Differential-drive joystick calculation

The Pico W combines the joystick coordinates to calculate the left and right motor commands.

The differential-drive equations are:

$$L_{\text{raw}} = y + x$$

$$R_{\text{raw}} = y - x$$

If either result exceeds the permitted range, both values are normalised using the largest absolute value:

$$M = \max\left(|L_{\text{raw}}|, |R_{\text{raw}}|\right)$$

When $M > 1$, the values are divided by M.

The resulting values are scaled to the PWM range.

The sign of each motor command determines its direction, while its absolute value determines the PWM magnitude.

This allows proportional steering. For example, when the joystick is pushed forward while slightly to the right, the two motors receive different commands, producing a curved path.

2.2.7 Dead zone and command updates

A small dead zone is used around the centre of the joystick to avoid unwanted movement caused by minor touch-position changes.

The dead-zone threshold is:

$$r < 0.05$$

where:

$$r = \sqrt{x^2 + y^2}$$

When the joystick radius is below the threshold, the controller stops the motors.

The browser sends joystick updates every 20 milliseconds, corresponding to a nominal update frequency of:

$$f = \frac{1}{0.020} = 50 \text{ Hz}$$

This update interval is intended to provide responsive control while keeping the browser and controller synchronised during movement.

2.2.8 Motor connections and PWM

The Pico W uses the following GPIO mapping for the L293D motor driver.

Function	Pico W GPIO	Purpose
Left enable / PWM	10	Left motor speed
Left direction 1	11	Left motor direction
Left direction 2	12	Left motor direction
Right direction 1	13	Right motor direction
Right direction 2	14	Right motor direction
Right enable / PWM	15	Right motor speed

The right motor uses an inversion setting to account for its physical orientation.

The PWM range is configured from 0 to 255, with a frequency of 1000 Hz.

The practical starting threshold of the motors is also important. A PWM command that is too small may not overcome the motor's starting friction.

A PWM value around 110 was identified as a practical starting threshold during testing. The exact minimum value should remain consistent with the final firmware revision.

2.2.9 Circuit diagram

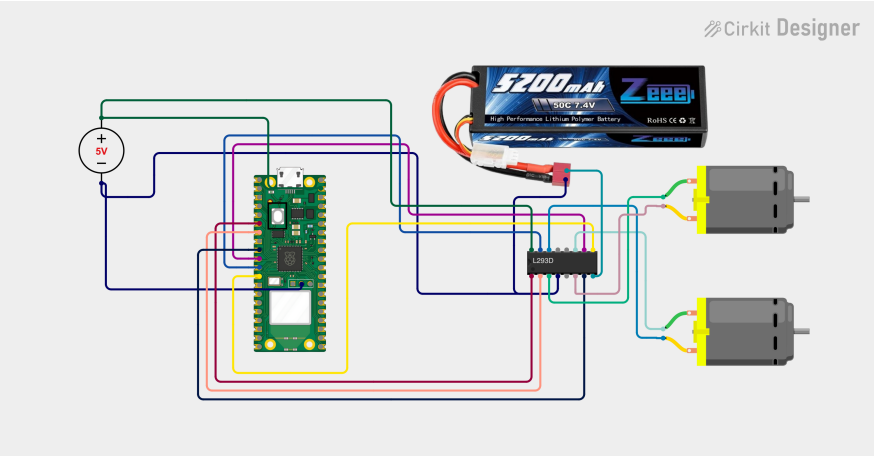


Figure 5: Raspberry Pi Pico W and L293D circuit diagram.

2.2.10 Command timeout and stopping behaviour

A remote-control system should not continue moving indefinitely if the control connection is interrupted.

The browser sends a stop request when the user releases or cancels the joystick interaction.

The firmware also contains a watchdog mechanism. If the controller does not receive a new movement command within the configured timeout, it stops the motors.

The watchdog timeout in the documented firmware is:

$$T_{\text{watchdog}} = 250 \text{ ms}$$

The browser-side stop request is the normal stopping mechanism. The watchdog provides a backup if the browser stops sending updates because of a connection interruption or another unexpected event.

These two mechanisms serve different purposes and work together to improve stopping reliability.

2.2.11 Serial monitoring and telemetry

The firmware provides serial output to help observe the controller's response to joystick inputs.

The telemetry includes:

- Joystick X and Y coordinates.
- Joystick radius and direction.
- Left and right motor directions.
- Signed motor commands.
- PWM magnitudes and duty-cycle percentages.
- Watchdog state.

The serial monitor operates at 115200 baud. This information is useful for observing the internal control calculations and troubleshooting the motor-control system.

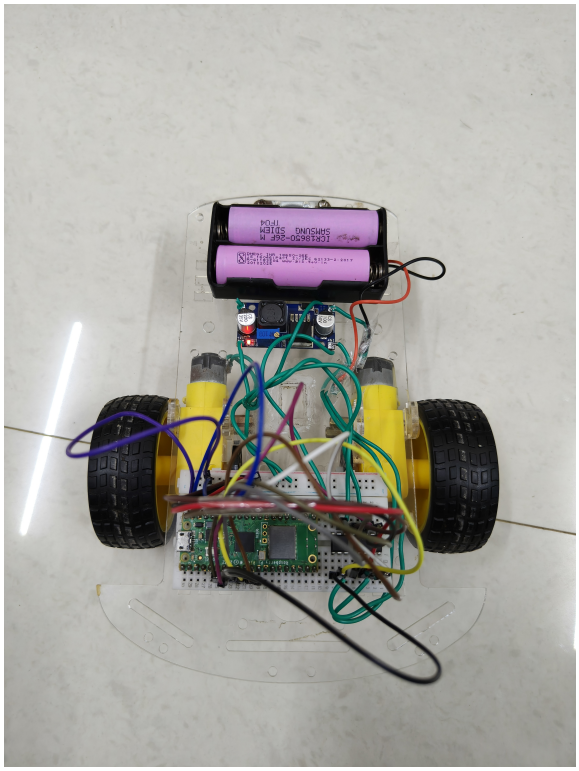


Figure 6: Assembled Raspberry Pi Pico W local Wi-Fi rover.

2.2.12 Controller transition and design limitation

The development of the local Wi-Fi model involved a change of controller.

The ESP32 DevKit had worked properly in the initial Bluetooth model. However, its later Wi-Fi operation was not sufficiently reliable during testing. We suspected that the power arrangement might have contributed to the problem, but the exact cause was not confirmed.

We then attempted to use a NodeMCU board, but it did not work as expected for the intended implementation.

We therefore moved to the Raspberry Pi Pico W for the local Wi-Fi joystick model.

The Pico W successfully provided the local access-point architecture required for browser-based control.

However, this local access-point arrangement is different from connecting directly to the IITH campus network using WPA2-Enterprise authentication.

2.2.13 Model outcome

The local Wi-Fi model established a browser-based control approach using the Raspberry Pi Pico W.

Its main features include:

- A local Wi-Fi access point named `UGV`.
- An HTTP server hosted by the Pico W.
- A browser-based joystick interface.
- Proportional differential-drive control.

- PWM-based motor speed control.
- Multi-device access for up to four clients.
- A STOP request and watchdog timeout.
- Serial telemetry for observing control values.

The model provides a flexible local control architecture in which multiple devices can connect to the same rover without requiring an external router or internet connection.

2.2.14 Source code and demonstration

The complete firmware is maintained separately from this report.

Source-code path:

```
codes/wifi_pico_w/main.ino
```

Circuit diagram link:

Open Raspberry Pi Pico W circuit in Cirkit Designer

Demonstration video:

Watch Pico W local Wi-Fi joystick demonstration

2.3 Model 3: ESP32-CAM Campus Camera Rover

2.3.1 Objective and system overview

The third model was intended to extend the rover's capabilities by combining remote operation with a camera-based view of its surroundings.

An ESP32-CAM board was used for this model. The intended architecture involved connecting the board to the IITH campus Wi-Fi network and using the camera to provide a live video feed.

The operator would be able to access the rover over the network and observe the camera feed while issuing movement commands.

2.3.2 Campus Wi-Fi connection

The ESP32-CAM was successfully connected to the IITH campus Wi-Fi network using WPA2-Enterprise authentication.

This confirmed that the campus-network connection could be established with the ESP32-CAM.

The remaining difficulty was associated with the camera hardware and its interfacing.

2.3.3 Camera interfacing

The camera module was intended to provide a live video stream from the rover.

However, during setup, we encountered a camera hardware or interfacing problem that prevented successful camera initialisation.

As a result, we could not complete the live-streaming implementation or demonstrate the intended combined camera and motor-control system.

The camera problem should be treated separately from the campus Wi-Fi connection, since the network connection itself was successfully established.

2.3.4 Circuit diagram

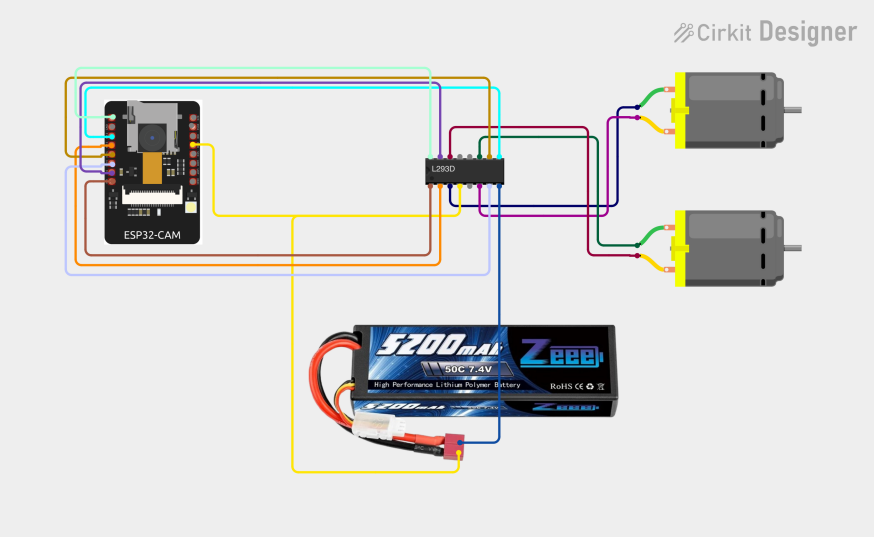


Figure 7: ESP32-CAM rover circuit diagram.

2.3.5 Prescribed motor-driver GPIO mapping

The circuit reference specifies the following ESP32-CAM connections to the L293D:

Function	ESP32 GPIO	L293D pin
Left direction input 1	13	2
Left direction input 2	14	7
Right direction input 1	15	10
Right direction input 2	2	15
Left enable / PWM (EN1)	4	1
Right enable / PWM (EN2)	12	9

The L293D logic supply connects to pin 16, while the motor supply connects separately to pin 8. All circuit grounds must be common. The motor battery positive must not be connected to the ESP32-CAM 5 V pin. This is the prescribed mapping from the circuit reference; its compatibility must be checked against the exact ESP32-CAM board and camera configuration because some GPIOs may be shared with camera, flash, or SD functions.

2.3.6 Intended operation

The intended operating sequence was:

1. Power the ESP32-CAM and initialise the board.
2. Connect the board to the IITH campus Wi-Fi network using WPA2-Enterprise authentication.
3. Initialise the camera module.
4. Start the camera server or streaming interface.
5. Access the rover through an authorised device on the network.

6. Use the camera feed to observe the surroundings while controlling the rover.

The camera initialisation problem prevented completion of this sequence.

2.3.7 Further development

The next step would be to verify the camera module independently and confirm that it can initialise successfully.

After that, the live video stream should be tested separately from the motor-control system. Once both the camera and network functions operate reliably, they can be integrated into a single remote-control interface.

Further testing should examine video delay, connection stability, and the effect of motor operation on the camera feed.

2.3.8 Model outcome

The ESP32-CAM model established the intended direction for a campus-network-connected camera rover.

The WPA2-Enterprise campus Wi-Fi connection was successful. However, the camera interfacing problem prevented completion of the live video stream.

The model is therefore recorded as an attempted implementation with an incomplete camera subsystem. A successful live camera feed or completed camera-based remote-control demonstration is not claimed.

2.3.9 Source code and demonstration

The camera firmware is maintained separately from this report.

Source-code path:

```
codes/esp32_cam/main.ino
```

Circuit diagram link:

Open ESP32-CAM rover circuit in Cirkit Designer

No demonstration video is included because a successful camera-streaming demonstration was not completed.

2.4 Research Models

2.4.1 Dynamic Node Hopping

Research objective

Dynamic Node Hopping explores a possible way to extend the communication range of the UGV by using intermediate wireless relay nodes.

In a direct communication system, the operator's device communicates with the rover through a single wireless link. The useful range is therefore limited by the quality and availability of that connection.

Dynamic Node Hopping proposes intermediate nodes that forward commands and telemetry between the operator and the rover.

2.4.2 Proposed architecture

The proposed arrangement consists of three main elements:

- **Operator node:** The phone or computer used to issue movement commands and receive information from the rover.
- **Intermediate relay nodes:** Wireless nodes that receive and forward packets between other nodes.
- **UGV node:** The final node that receives movement commands and sends telemetry back through the network.

A simplified communication path is:

$$\text{Operator} \longrightarrow \text{Relay 1} \longrightarrow \text{Relay 2} \longrightarrow \text{UGV}$$

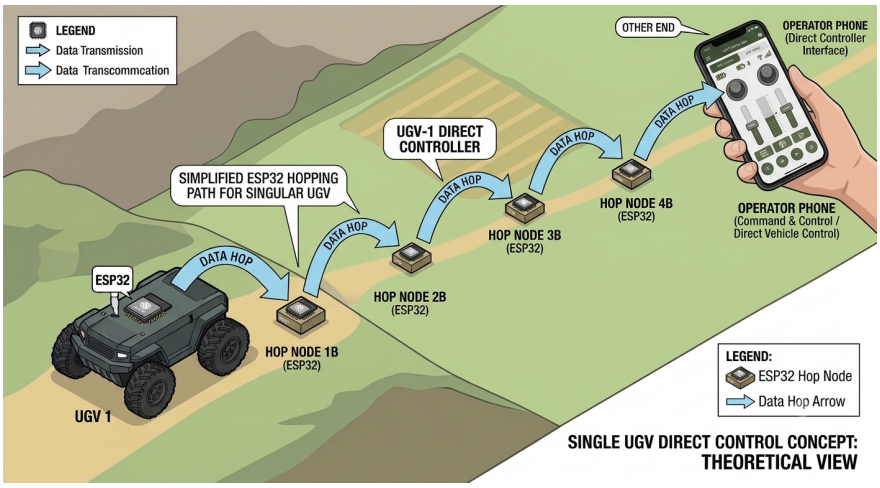


Figure 8: Proposed Dynamic Node Hopping architecture.

2.4.3 Dynamic route selection

A key part of the concept is the ability to select a suitable communication route.

If the rover moves away from one relay node or a wireless connection becomes weak, the system could attempt to select another available path.

Possible factors for route selection include:

- Signal quality between neighbouring nodes.
- Communication delay.
- Packet loss.
- Availability of relay nodes.
- Number of hops required to reach the rover.

A route with more relay nodes may extend the communication range, but it can also introduce additional delay.

2.4.4 Communication reliability and safe stopping

Remote motor control requires predictable communication.

If a relay node becomes unavailable, the system must detect the communication failure and respond appropriately.

The proposed design should include:

- Detection of missing or failed relay links.
- Monitoring of packet delivery and delay.
- Route recovery when an alternative path exists.
- A timeout mechanism for missing control commands.
- A safe stop condition when communication is lost.

2.4.5 Future research and testing

The concept requires further investigation before a physical implementation can be claimed.

Some of the questions to be studied are:

1. How should relay nodes discover neighbouring nodes?
2. How should the system select a route to the rover?
3. How much delay is introduced by each additional hop?
4. How does packet loss affect joystick control?
5. How quickly can the network recover after a relay node becomes unavailable?
6. How reliably can the rover stop when the communication path is interrupted?

A future experimental setup could measure packet loss, command delay, route recovery time, and stopping response under different network conditions.

2.4.6 Research status

Dynamic Node Hopping is a proposed research direction for extending the communication range of the UGV.

A complete multi-node firmware system and physical relay network were not implemented or tested during the work recorded in this document.

No measured range improvement, packet-loss result, or routing performance is therefore claimed.

The concept remains a possible future extension of the wireless control system.

3 Conclusion

The project explored different approaches to wireless control and remote operation of an Unmanned Ground Vehicle.

The first model used an ESP32 DevKit with Bluetooth communication through the Dabble application. This provided the initial working approach for wireless motor control.

The second model used a Raspberry Pi Pico W to create a local Wi-Fi access point and host a browser-based joystick interface. It introduced HTTP communication, proportional joystick control, differential-drive calculations, and watchdog-based stopping behaviour. Its access-point architecture also supports up to four connected Wi-Fi clients simultaneously.

The third model explored a camera-assisted rover using an ESP32-CAM connected to the IITH campus Wi-Fi network. The WPA2-Enterprise connection was successful, but a camera interfacing problem prevented completion of the live video stream.

Finally, Dynamic Node Hopping was identified as a possible research direction for extending the communication range through intermediate relay nodes. It remains a conceptual proposal rather than a completed implementation.

The practical work highlighted the importance of reliable wiring, suitable power arrangements, correct motor-driver connections, and dependable communication between the operator and the rover.